Seasonal Malaria Chemoprevention (SMC) Project Performance Report

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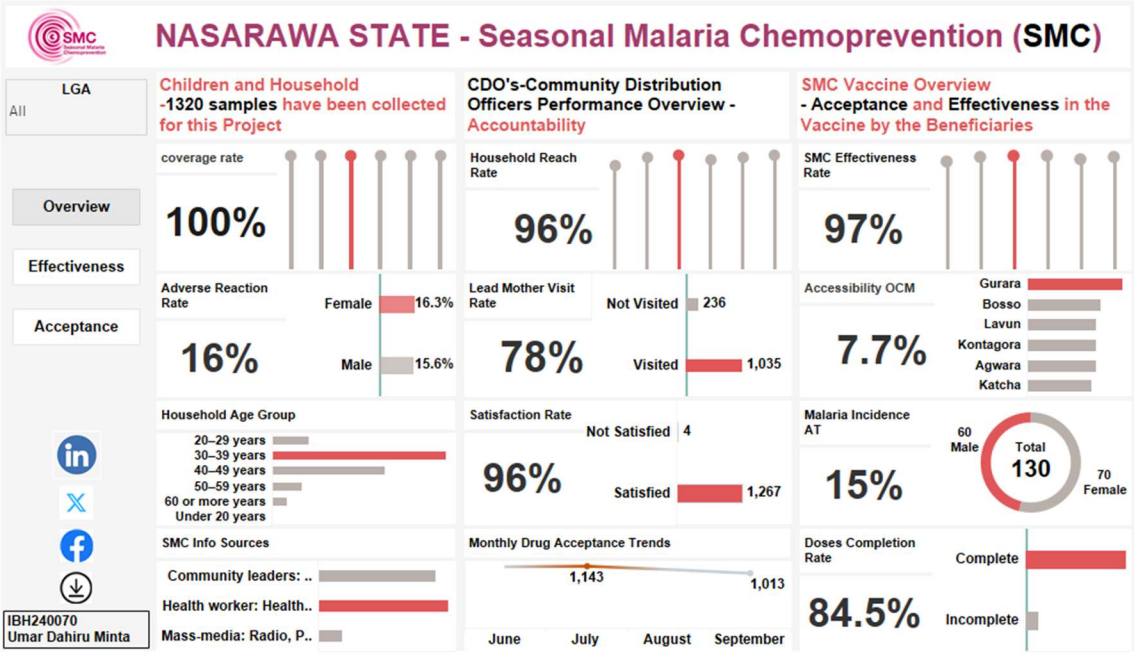
Executive Summary

This report presents the performance analysis of the Seasonal Malaria Chemoprevention (SMC) program using data collected from 1,320 children and households. The evaluation spans across key indicators including vaccine effectiveness, adverse reactions, program acceptance, and community reach. Results show excellent vaccine coverage and effectiveness, moderate adverse reaction rates, and high overall satisfaction among beneficiaries.



SMC VACCINE EFFECTIVENESS DASHBOARD

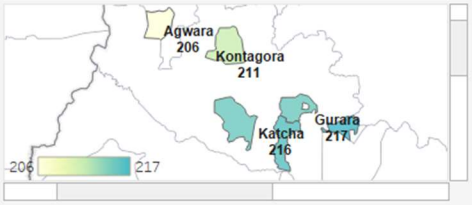
Vaccine Completion Rate	Adverse Reaction Rate	Doses Completion Rate	Malaria Incidence AT	SMC Effectiveness Rate
100%	16%	84.5%	15%	97%

Overview

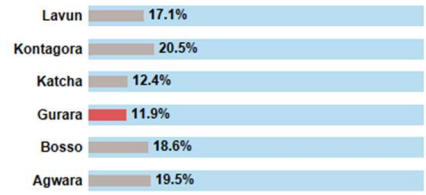
Effectiveness

Acceptance

SMC Confidence Rate/Region



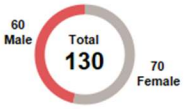
Adverse Reaction Rate/LGA



Adverse Reaction/LGA

LGA	D	D..	D..	D..	F	F..	I	I/F	I/..	LA	NR	N..	S	S..	SV	S..	S..	S..	S..	S..	YE	Y..	Y..	Y..
Agwara	2			1	4	1	1	1		3		11	6	3	1			4	1				1	1
Bosso	4				7		2		1			5	6		10		1	3						
Gurara					9					2	1	4	2	1	2	1		3						
Katcha	2		1		4					2		4	4		7						1			1
Kontagora	1	1			8	1				1		9	6	2	6	1		4		2		1		
Lavun	1				11	1				5		6	3		7			1						1

Malaria +/Gender



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SMC ACCEPTANCE RATE DASHBOARD

Vaccine Completion Rate	SMC Effectiveness Rate	Doses Completion Rate	Satisfaction Rate
100%	97%	84.5%	96%

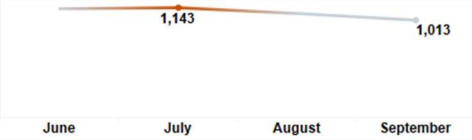
Overview

Effectiveness

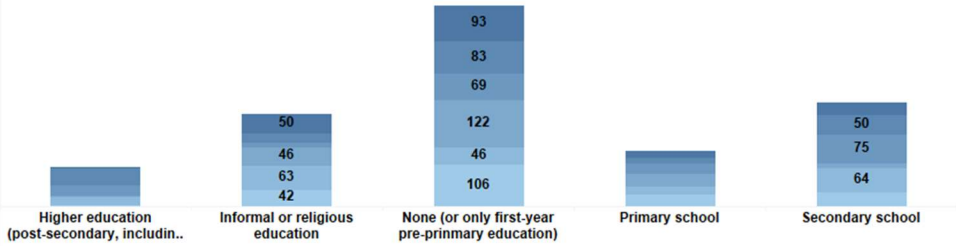
Acceptance

Satisfaction Rate/ LGA

Monthly Drug Acceptance Trends



House Hold / LGA (Acceptance rate)



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1. SMC Vaccine Effectiveness

Definitions:

- PPV (Proportion of Population Vaccinated With SMC): 90.2%

- PCV (Proportion of Cases Vaccinated With SMC): 9.8%

Formula for Vaccine Effectiveness (VE):

$VE = (PPV - PCV) / (PPV \times (1 - PCV)) \times 100$

Substituting the given values:

$VE = (0.902 - 0.098) / (0.902 \times (1 - 0.098)) \times 100$

$VE = 0.804 / (0.902 \times 0.902) \times 100$

$VE = 0.804 / 0.813604 \times 100 \approx 98.82\%$

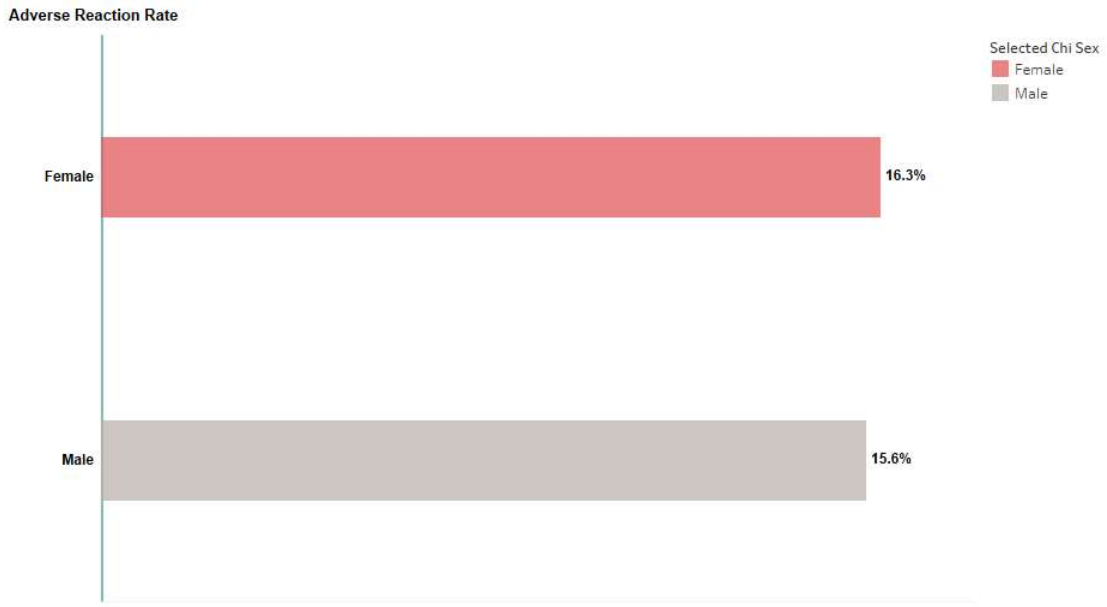
Observation:

the estimated effectiveness of the Seasonal Malaria Chemoprevention (SMC) vaccine is approximately 98.82%. This high level of effectiveness indicates a strong protective impact of the SMC intervention among the target population.

2. Key Performance Indicators (KPIs)

Metric	Value
Vaccine Completion Rate	100%
Doses Completion Rate	84.5%
SMC Effectiveness Rate	97%
Adverse Reaction Rate	16%
Malaria Incidence Rate (AT)	15%
Overall Satisfaction Rate	96%
Household Reach Rate	96%

3. Adverse Reaction Analysis



Sum of ARR for each Selected Chi Sex. Color shows details about Selected Chi Sex. The marks are labeled by ARR %. The data is filtered on LGA, which keeps multiple members.

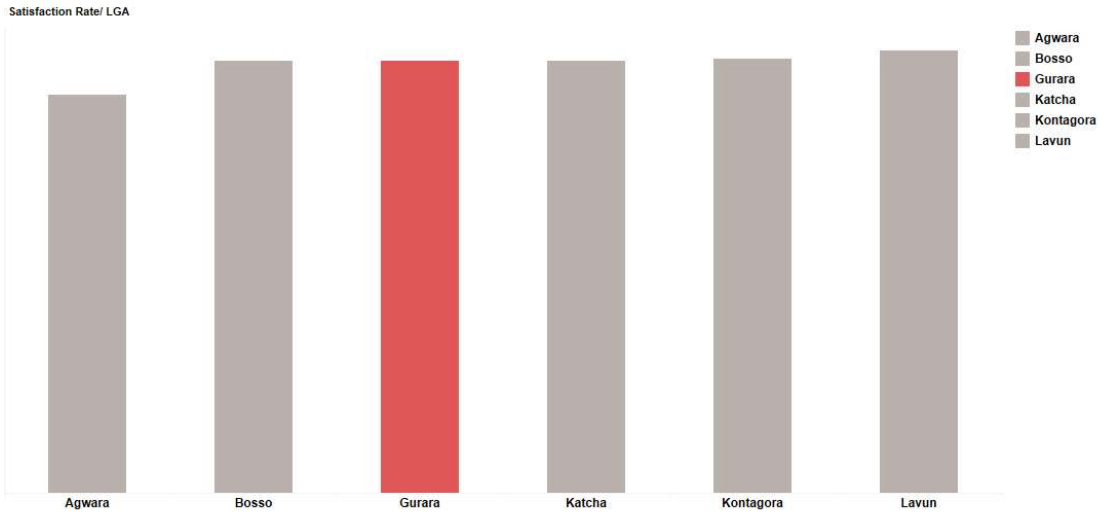
- Overall Adverse Reaction Rate: 16%
- By Gender:
  - Female: 16.3%
  - Male: 15.6%
- By LGA:

Count of Calc Month Cycle Name and sum of ARR for each LGA. The marks are labeled by sum of ARR. For pane Sum of ARR: Color shows details about LGA. The marks are labeled by % of Total ARR and sum of ARR. For pane Count of Calc Month Cycle Name: Color shows details about count of Calc Month Cycle Name. The view is filtered on LGA, which keeps multiple members.

- Highest in Kontagora (20.5%), Agwara (19.5%), and Bosso (18.6%)
- Lowest in Gurara (11.9%) and Katcha (12.4%)

Observation: The data suggests that although the adverse reaction rate is within acceptable limits, targeted follow-ups in LGAs like Kontagora and Agwara may help improve outcomes further.

4. Community Acceptance and Confidence

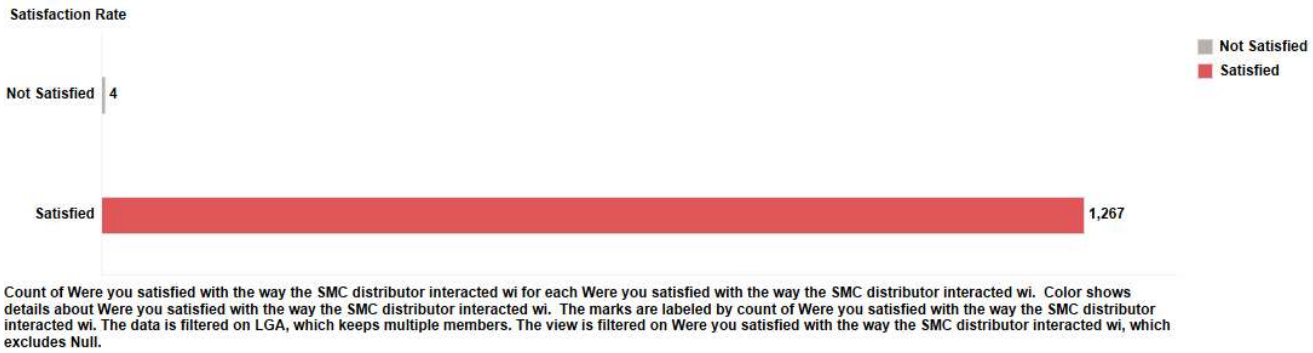


Sum of PSR for each LGA. Color shows details about LGA. The view is filtered on LGA, which keeps multiple members.

- Satisfaction Rate per LGA: Uniformly high across all LGAs, with Gurara leading.
- Drug Acceptance Trends: A decline observed from June (1,143) to September (1,013).
- Household Acceptance by Education Level:
  - Highest in households with no or only pre-primary education.

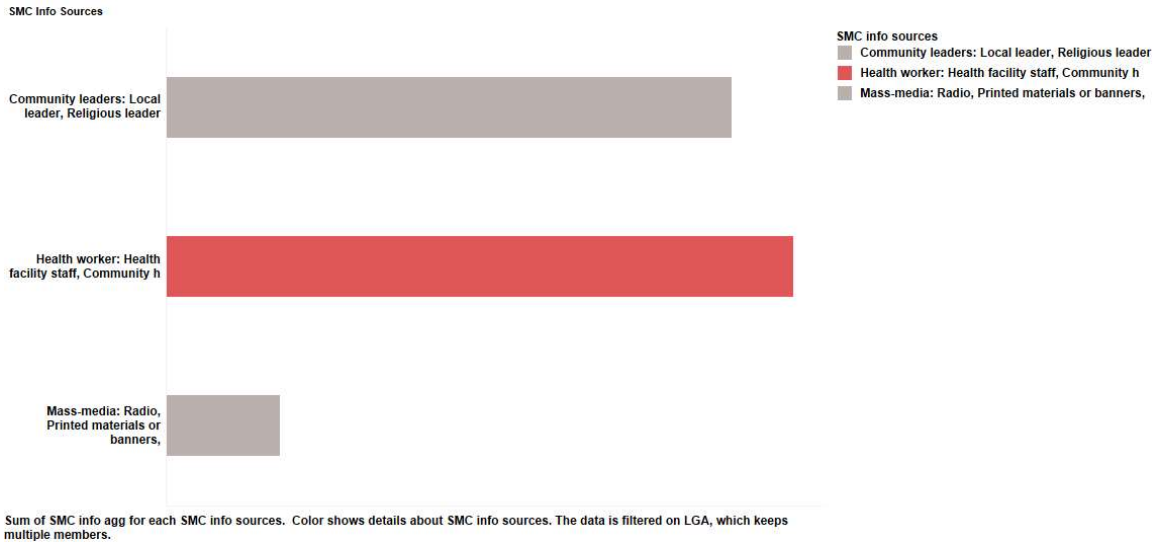
- Moderate in primary and secondary school categories.
- Lower in households with higher education.

5. Accessibility and Outreach



- Community Distribution Officers (CDOs): 132 officers, including data collectors and 12 supervisors.
- Lead Mother Visits:
  - Visited: 1,035
  - Not Visited: 236
- Accessibility Challenges (OCM): 7.7% reported difficulties accessing services.

6. Demographics and Source of Information



- Age Group with Most Respondents: 30–39 years
- Information Channels:
  - Community leaders and health workers were the most effective sources which indicate that mass media is less effective since the target communities are fragile. Spend less in mass media and focus in the 2 effective way which they are coast effective.

7. Gender and Malaria Incidence

- Malaria Positive Cases: 130
  - Female: 70
  - Male: 60

This suggests a slightly higher burden among females, although the difference is minimal.

Recommendations

1. Improve Dose Completion: Target the 15.5% of children who did not complete doses.
2. Address High Adverse Reaction LGAs: Monitor LGAs like Kontagora and Agwara.
3. Stabilize Acceptance Trends: Investigate drop in drug acceptance and improve community engagement.
4. Targeted Education Campaigns: Tailor messages for households with higher education.
5. Leverage Health Workers & Media: Expand use of trusted information channels.

Conclusion

The SMC program has demonstrated strong effectiveness and acceptance, with full vaccine completion and a very high satisfaction rate. Minor concerns around dose completion and adverse reactions in specific LGAs provide opportunities for improvement. With data-driven adjustments and targeted community engagement, the program can achieve even greater impact.

About: